

Annexure 3: Energy Conservation

Annexure 3 ascertains what steps a company takes to conserve energy and what kind of technology is used to achieve the same. It also talks about the foreign exchange earnings of the company.

 Corporate Overview Statutory Reports Financial Statements	
ANNEXURE-03 TO DIRECTORS' REPORT 2020-21 ENERGY CONSERVATION, TECHNOLOGY ABSORPTION AND FOREIGN EXCHANGE EARNINGS AND OUTGO.	
(A) CONSERVATION OF ENERGY-	
(i) The steps taken or impact on conservation of energy:	
Steps Taken:	
1) Variable Frequency Drive (VFD) provided for ABC HP Pump Kiln- 2 Sponge Iron Division (SID) Phase-1.	9) Change in Starter Method from Star Delta Motor Starter to Variable Frequency Drive.
2) Variable Frequency Drive provided for ABC Fan Kiln- 2 Sponge Iron Division Phase-1.	The impact of above measures:-
3) Replacement of High Pressure Sodium Vapour lights with LED lights in Sponge Iron Division.	1) By providing VFD for ABC HP Pump Kiln-2 SID Ph-1 there was a saving of 188100 units of Power annually;
4) Replacement of Water Cooled AC to Air Cooled AC in Sponge Iron Division Phase-2Control Room.	2) By providing VFD for ABC Fan Kiln-2 SID Ph-1 there was saving of 189420 units of Power annually.
5) Started Consuming low Gangue DRI Pellet instead of high Gangue DRI Pellet.	3) By replacing of HPSV lights with LED lights in SID there was saving of 103114 units per year.
6) Installation of Sonic soot blower in Waste Heat Recovery Boiler-1 & 4.	4) By replacement of Water Cooled AC to Air Cooled AC in SID-2 Control Room there was saving of 153578 units per year.
7) Replacement of conventional lighting with LED lights.	5) Consuming Low Gangue DRI Pellet instead of High Gangue DRI Pellet reduced the heat timings due to which Power Consumption Reduced.
8) Change in Motor Pump Rating from 75KW Motor to 45 KW Motor.	6) Installation of Sonic soot blower reduced Electricity and Coal consumption.
	7) Replacement of conventional lighting with LED lights saved electricity 18000KWH per year.
(ii) The steps taken by the company for utilizing alternate sources of energy	
	None
(iii) Capital investment on energy conservation equipments	
	₹ In Lacs
1. VFD provided for ABC HP Pump Kiln- 2 SID Ph-1.	8.00
2. VFD provided for ABC Fan Kiln- 2 SID Ph-1.	2.50
3. Replacement of HPSV lights with LED lights in SID Ph-1 & 2.	4.40
4. Replacement of Water Cooled AC to Air Cooled AC in SID-2 Control Room.	11.00
5. Installation of Sonic soot blower.	12.00
6. Replacement of CT make up pumps.	0.20
7. VFD Purchased 2 nos. 55KW Schneider make	2.35
TOTAL INVESTMENTS	40.45
SAVINGS:	
1. Savings by providing VFD for ABC HP Pump Kiln-2 Ph-1 (@ ₹5/- Per unit)	9.40
2. Savings by providing VFD for ABC Fan Kiln-1 SID Ph-1 (@ ₹5/- Per unit)	9.40
3. Savings by Replacement of HPSV lights with LED lights in SID (@ ₹5/- Per unit).	5.10
4. Savings by Replacement of Water Cooled AC to Air Cooled AC in SID-2 Control Room.	7.60
5. Savings in electricity by Started Consuming Low Gangue DRI Pellet instead of High Gangue DRI Pellet as compared to previous FY 2019-2020 :-	451.00
6. Savings on coal 40MT (@ ₹ 350/MT) and electricity saving 32200 KWH.	3.00
7. Savings in electricity by replacement of conventional lighting to LED is 16000 KWH per year (@ ₹ 5 /- Per unit).	0.80
8. Savings in KWH consumptions per day by 429 units	6.18
TOTAL SAVINGS	492.78
(C) FOREIGN EXCHANGE EARNINGS AND OUTGO	
The Foreign Exchange earned in terms of actual inflows during the year	₹125284.62 Lacs
Foreign Exchange outgo during the year in terms of actual outflows.	₹25114.94 Lacs